An Exploration

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Executive Summary

Data Cleaning

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(zoo)
```

```
Attaching package: 'zoo'

The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

```r
library(stringr)
library(gt)

i_res_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/CSV/res_survey.csv", n_max

questions <- as.character(i_res_headers[1,])

answer_choices <- as.character(i_res_headers[2,])

questions_filled <- na.locf(ifelse(questions == "" | is.na(questions), NA, questions))

clean_headers <- paste(questions_filled, answer_choices, sep = "_")

i_res_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/CSV/res_survey.csv", skip = 2,
i_res <- i_res_raw %>%
  select(c(10:ncol(i_res_raw)))

i_res <- i_res %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )
i_res <- i_res %>%
  select(!"Which departments are primarily responsible for management of the following proce
i_res_q <- i_res %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
  mutate(Q3 = do.call(coalesce, pick(6:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:13))) %>%
  mutate(Q5 = do.call(coalesce, pick(14:15))) %>%
  mutate(Q6 = do.call(coalesce, pick(16:18))) %>%
  mutate(Q7a = do.call(coalesce, pick(19:21))) %>%
  mutate(Q7b = do.call(coalesce, pick(22:24))) %>%
  mutate(Q7c = do.call(coalesce, pick(25:27))) %>%
  mutate(Q7d = do.call(coalesce, pick(28:30))) %>%
  mutate(Q8a = do.call(coalesce, pick(31:35))) %>%
  mutate(Q8b = do.call(coalesce, pick(36:40))) %>%
  mutate(Q8c = do.call(coalesce, pick(41:45))) %>%
  mutate(Q8d = do.call(coalesce, pick(46:50))) %>%
  mutate(Q9 = do.call(coalesce, pick(51:57))) %>%
```

```
  mutate(Q10a = do.call(coalesce, pick(58:64))) %>%
  mutate(Q10b = do.call(coalesce, pick(65:71))) %>%
  mutate(Q10c = do.call(coalesce, pick(72:78))) %>%
  mutate(Q10d = do.call(coalesce, pick(79:85))) %>%
  mutate(Q10e = do.call(coalesce, pick(86:92))) %>%
  mutate(Q11 = do.call(coalesce, pick(93:94))) %>%
  mutate(Q12 = do.call(coalesce, pick(95:97))) %>%
  mutate(Q13 = do.call(coalesce, pick(98:102))) %>%
  mutate(Q14 = do.call(coalesce, pick(103:105))) %>%
  mutate(Q15 = do.call(coalesce, pick(106:109)))

i_res_q$role <- "Resident"

i_res_coalesced <- i_res_q %>%
  select(110:135)
```

```
i_dir_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/i_directors.csv", n_max = 2

questions_dir <- as.character(i_dir_headers[1,])

answer_choices_dir <- as.character(i_dir_headers[2,])

questions_filled_dir <- na.locf(ifelse(questions_dir == "" | is.na(questions_dir), NA, questi

clean_headers_dir <- paste(questions_filled_dir, answer_choices_dir, sep = "_")

i_dir_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/i_directors.csv", skip = 2, col
i_dir <- i_dir_raw %>%
  select(c(10:ncol(i_dir_raw)))

i_dir <- i_dir %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )
i_dir <- i_dir %>%
  select(!"Which departments are primarily responsible for management of the following proced
i_dir_q <- i_dir %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
```

```
  mutate(Q3 = do.call(coalesce, pick(6:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:13))) %>%
  # mutate(Q5 = do.call(coalesce, pick(14:25))) %>%
  mutate(Q6 = do.call(coalesce, pick(26:30))) %>%
  # mutate(Q7a = do.call(coalesce, pick(31:37))) %>%
  mutate(Q8 = do.call(coalesce, pick(38:42))) %>%
  mutate(Q9 = do.call(coalesce, pick(43:45))) %>%
  mutate(Q10a = do.call(coalesce, pick(46:48))) %>%
  mutate(Q10b = do.call(coalesce, pick(49:51))) %>%
  mutate(Q10c = do.call(coalesce, pick(52:54))) %>%
  mutate(Q10d = do.call(coalesce, pick(55:57))) %>%
  mutate(Q11a = do.call(coalesce, pick(58:62))) %>%
  mutate(Q11b = do.call(coalesce, pick(63:67))) %>%
  mutate(Q11c = do.call(coalesce, pick(68:72))) %>%
  mutate(Q11d = do.call(coalesce, pick(73:77))) %>%
  mutate(Q12 = do.call(coalesce, pick(78:79))) %>%
  mutate(Q13 = do.call(coalesce, pick(80:86))) %>%
  mutate(Q14 = do.call(coalesce, pick(87:90))) %>%
  mutate(Q15 = do.call(coalesce, pick(91:94))) %>%
  mutate(Q16a = do.call(coalesce, pick(95:101))) %>%
  mutate(Q16b = do.call(coalesce, pick(102:108))) %>%
  mutate(Q16c = do.call(coalesce, pick(109:115))) %>%
  mutate(Q16d = do.call(coalesce, pick(116:122))) %>%
  mutate(Q16e = do.call(coalesce, pick(123:129))) %>%
  mutate(Q17 = do.call(coalesce, pick(130:131))) %>%
  mutate(Q18 = do.call(coalesce, pick(132:134))) %>%
  mutate(Q19 = do.call(coalesce, pick(135:139))) %>%
  mutate(Q20 = do.call(coalesce, pick(140:142))) %>%
  mutate(Q21 = do.call(coalesce, pick(143:146)))

i_dir_coalesced_partial <- i_dir_q %>%
  select(147:150, 14:25, 151, 31:37, 152:175)

i_dir_coalesced <- i_dir_coalesced_partial %>%
  mutate(Q5 = pmap_chr(select(., 5:16), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q7 = pmap_chr(select(., 18:24), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:4, 49, 17, 50, 25:48)

i_dir_coalesced$role <- "Director"
i_dir_coalesced <- i_dir_coalesced %>%
  mutate(Q21_new = case_when(
    Q21 == "3-Jan" ~ "1-3",
```

```
    Q21 == "6-Apr" ~ "4-6",
    Q21 == "10-Jul" ~ "7-10",
    Q21 == ">10" ~ ">10"
  ))
```

Residents Analysis

```
# 2 with 7 and 8
# 2 with 11
# 2 with 11 and 12
question1s <- c("Q7a", "Q7b", "Q7c", "Q7d", "Q8a", "Q8b", "Q8c", "Q8d", "Q11", "Q12")

question2s <- c(rep("Q2", 10))

question_dic_res <- data.frame(question1s, question2s)
```

```
validation <- function(question1, question2, data){
  df <- data %>%
    transmute(
      response1 = .data[[question1]],
      response2 = .data[[question2]]
    ) %>%
    drop_na()

  comparison_res <- df %>%
    group_by(response1, response2) %>%
    summarize(counts = n(), .groups = "drop") %>%
    group_by(response2) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  table_comparison <- table(df$response2, df$response1)
  #table_hours
  test <- suppressWarnings(chisq.test(table_comparison))

  if (mean(test$expected <= 5) < 0.8 | any(test$expected < 1)){
    fish_test <- fisher.test(table_comparison, simulate.p.value = TRUE)
    if (fish_test$p.value <= 0.05){
      significant <- TRUE
    }
```

```
    else {
      significant <- FALSE
    }
    return(c("question1", "question2", round(fish_test$p.value, digits = 4), significant, "fi
  }
  else{
    if (test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
  }
  return(c("question1", "question2", round(test$p.value, digits = 4), significant, "chisq"))
}
```

```
res_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(question_dic_res))) {

  res_results[i, ] <- validation(
    question1 = question_dic_res$question1s[i],
    question2 = question_dic_res$question2s[i],
    data = i_res_coalesced
  )
}

res_results <- res_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = question_dic_res$question1s,
    Question2 = question_dic_res$question2s
  ) %>%
  arrange(Adj_P)
```

Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q8a	Q2	5e-04	TRUE	fisher	0.0017	TRUE
Q8b	Q2	5e-04	TRUE	fisher	0.0017	TRUE
Q8c	Q2	5e-04	TRUE	fisher	0.0017	TRUE
Q8d	Q2	0.0045	TRUE	fisher	0.0112	TRUE
Q12	Q2	0.1294	FALSE	fisher	0.2588	FALSE
Q7a	Q2	0.8636	FALSE	fisher	0.8636	FALSE
Q7b	Q2	0.7131	FALSE	fisher	0.8636	FALSE
Q7c	Q2	0.7851	FALSE	fisher	0.8636	FALSE
Q7d	Q2	0.5247	FALSE	fisher	0.8636	FALSE
Q11	Q2	0.6482	FALSE	fisher	0.8636	FALSE

```
gt(res_results)
```

Question 2 with Questions 11 and 12

```
library(modelsummary)
cleft_program <- i_res_coalesced %>%
  filter(Q2 != "Unsure") %>%
  mutate(cleft_response = as.factor(Q11),
         program = as.factor(Q12),
         cleft_center = as.factor(Q2)) %>%
  select(cleft_response, cleft_center, program) %>%
  drop_na()
cleft_glm <- glm(cleft_response ~ cleft_center + program, family = "binomial", data = cleft_
cleft_null <- glm(cleft_response ~ 1, family = "binomial", data = cleft_program)
anova(cleft_glm, cleft_null)
```

```
Analysis of Deviance Table

Model 1: cleft_response ~ cleft_center + program
Model 2: cleft_response ~ 1
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        73     89.195
2        76     99.769 -3  -10.573  0.01427 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	(1)
(Intercept)	1.059
	(0.865)
cleft_centerYes	0.713
	(0.758)
programHospital-based	−1.635
	(0.623)
programUniversity	0.236
	(1.290)
Num.Obs.	77
AIC	97.2
BIC	106.6
Log.Lik.	−44.598
F	2.950
RMSE	0.45

```
#logistic regression
#use model summary
# we notice signficance when comparing the model with program and cleft centers to the null m
```

The logistic regression summary

```
modelsummary(cleft_glm)
```

Residents & Directors Cross-Referenced

```
common_qs_res <- c("Q2", "Q3", "Q4", "Q6", "Q7a", "Q7b", "Q7c", "Q7d","Q8a","Q8b", "Q8c", "Q8
                   "role")
common_qs_dir <- c("Q2", "Q3", "Q4", "Q9", "Q10a", "Q10b", "Q10c", "Q10d", "Q11a","Q11b","Q11
                  "Q21_new",   "role")
q_dic <- data.frame(common_qs_res, common_qs_dir)
```

```
res_common <- i_res_coalesced %>%
  select(all_of(common_qs_res))

dir_common <- i_dir_coalesced %>%
  select(all_of(common_qs_dir)) %>%
  setNames(common_qs_res)
res_dir <- bind_rows(
  res_common,
  dir_common
)
```

Run the tests

```
res_dir_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(ncol(res_dir)-1)) {

  res_dir_results[i, ] <- validation(
    question1 = colnames(res_dir)[i],
    question2 = colnames(res_dir)[ncol(res_dir)],
    data = res_dir
  )
}

res_dir_results <- res_dir_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = colnames(res_dir)[1:(ncol(res_dir) -1)],
    Question2 = "role"
  ) %>% arrange(Adj_P)
gt(res_dir_results)
```

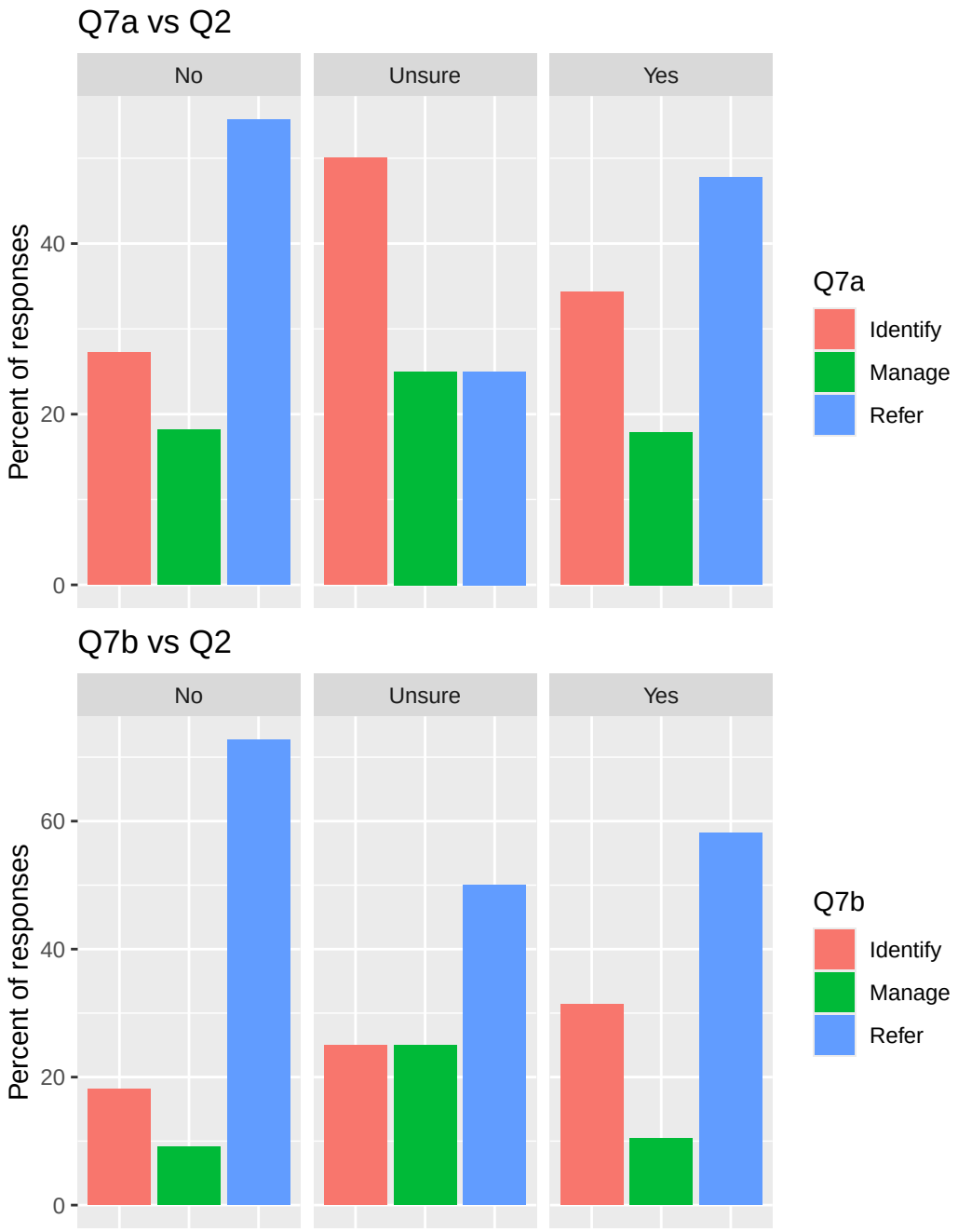
Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
Q8c	role	5e-04	TRUE	fisher	0.0115	TRUE
Q6	role	0.0035	TRUE	fisher	0.0402	TRUE
Q7b	role	0.0095	TRUE	fisher	0.0575	FALSE
Q8a	role	0.0125	TRUE	fisher	0.0575	FALSE
Q8b	role	0.01	TRUE	fisher	0.0575	FALSE
Q10a	role	0.0245	TRUE	fisher	0.0939	FALSE
Q9	role	0.0425	TRUE	fisher	0.1396	FALSE
Q7d	role	0.054	FALSE	fisher	0.1552	FALSE
Q7a	role	0.0895	FALSE	fisher	0.1871	FALSE
Q7c	role	0.074	FALSE	fisher	0.1871	FALSE
Q13	role	0.082	FALSE	fisher	0.1871	FALSE
Q5	role	0.1062	FALSE	fisher	0.2036	FALSE
Q8d	role	0.1449	FALSE	fisher	0.2564	FALSE
Q4	role	0.1744	FALSE	fisher	0.2865	FALSE
Q10e	role	0.2779	FALSE	fisher	0.4261	FALSE
Q10c	role	0.4263	FALSE	fisher	0.6128	FALSE
Q10b	role	0.4818	FALSE	fisher	0.6156	FALSE
Q12	role	0.4728	FALSE	fisher	0.6156	FALSE
Q10d	role	0.5847	FALSE	fisher	0.7078	FALSE
Q2	role	0.8431	FALSE	fisher	0.9696	FALSE
Q15	role	0.9055	FALSE	fisher	0.9917	FALSE
Q3	role	1	FALSE	fisher	1.0000	FALSE
Q14	role	1	FALSE	fisher	1.0000	FALSE

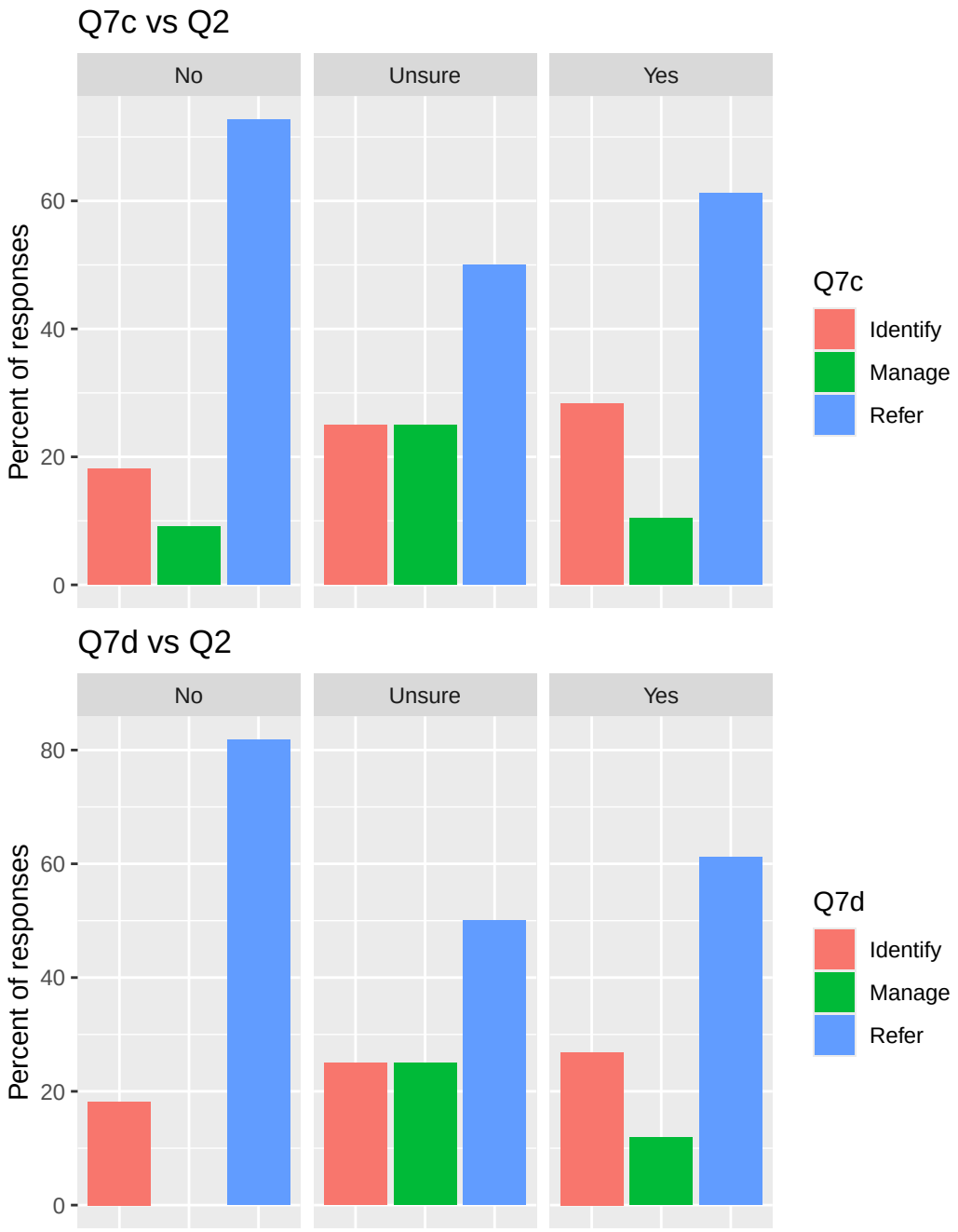
Visualization

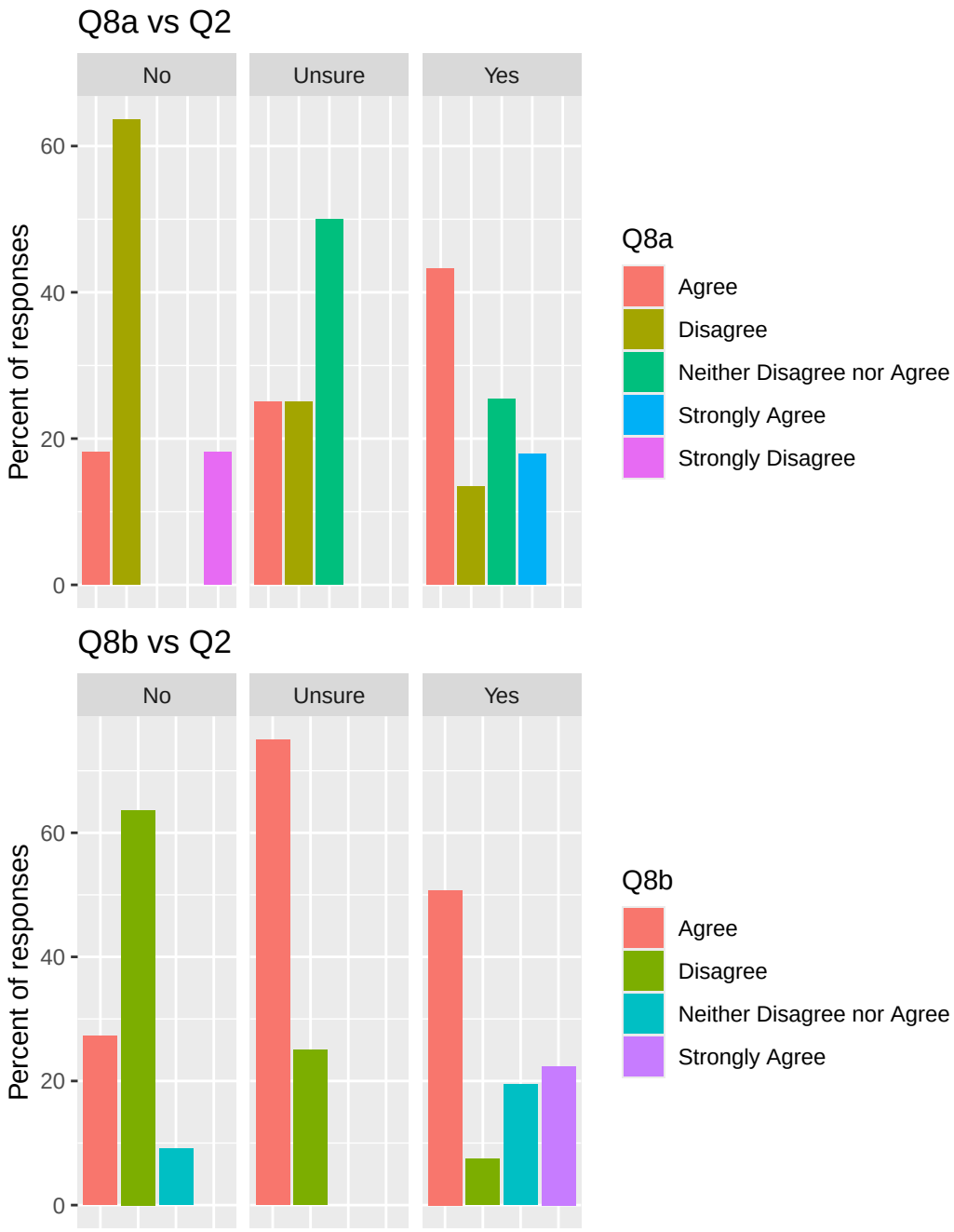
```
stack_bar <- function(q_reference, q_group, data){
  bar_df <- data %>%
    group_by(.data[[q_reference]], .data[[q_group]]) %>%
    summarize(counts = n(), .groups = "drop") %>%
    drop_na() %>%
    group_by(.data[[q_group]]) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

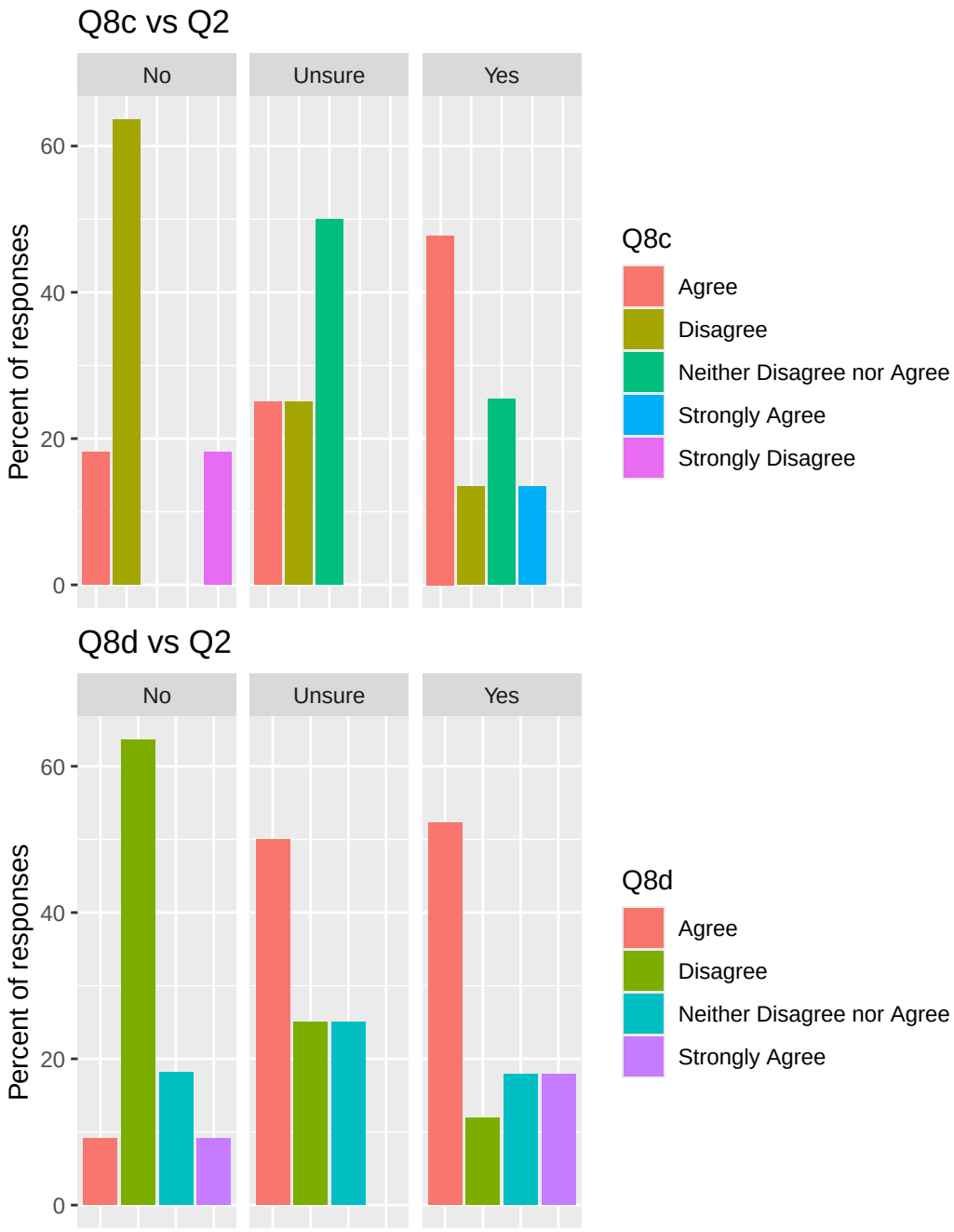
  p <- ggplot(bar_df, aes(x = .data[[q_reference]], y = percent, fill = .data[[q_reference]]
    geom_col() +
    facet_wrap(vars(.data[[q_group]])) +
    labs(
      title = paste(q_reference, "vs", q_group, sep = " "),
      x = NULL,
      y = "Percent of responses"
    ) +
    theme(
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank()
    )
  return(p)
}
```

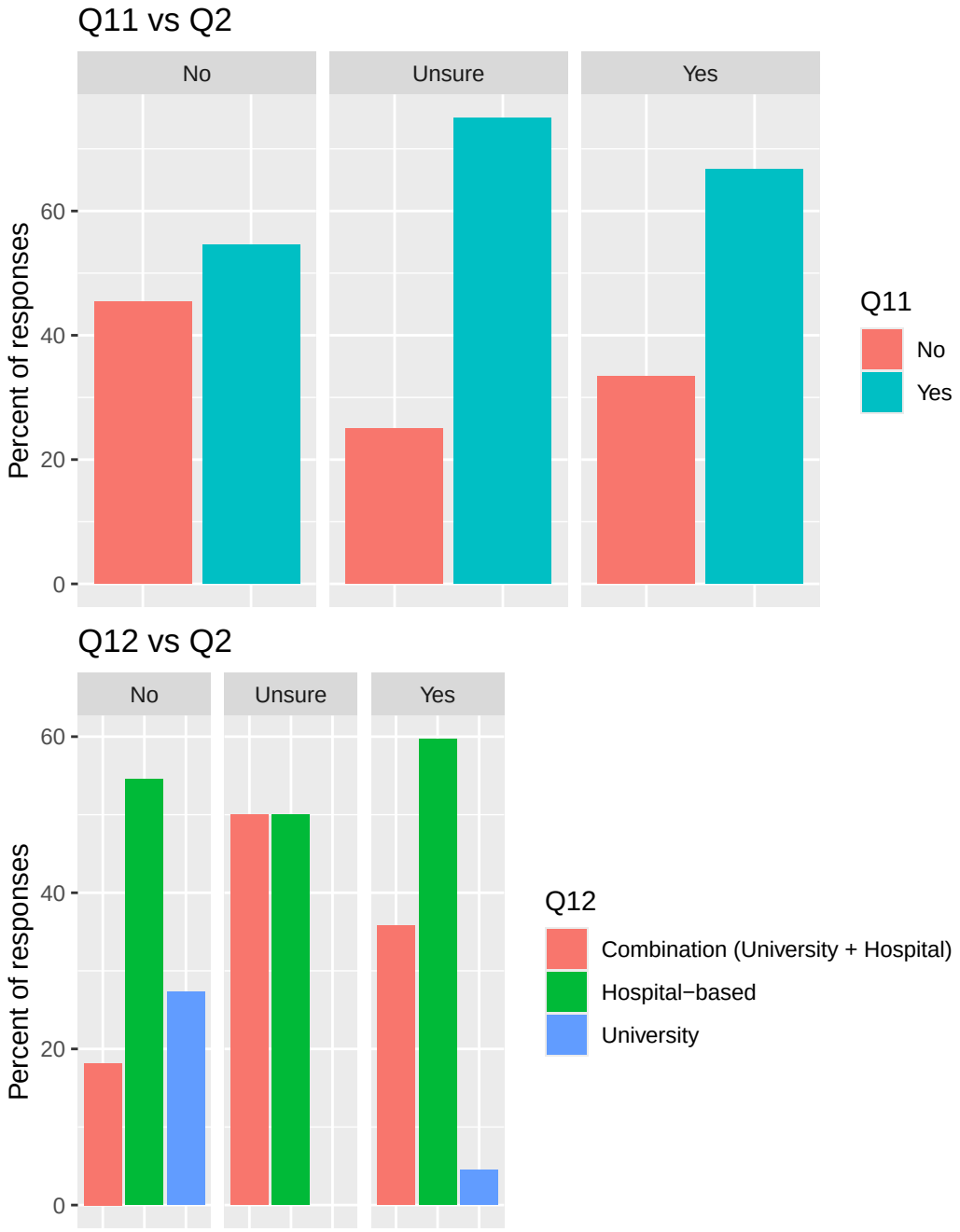
```
plots_res <- pmap(question_dic_res,
                  function(question1s, question2s){
                    stack_bar(
                      q_reference = question1s,
                      q_group = question2s,
                      data = i_res_coalesced
                    )
                  })
walk(plots_res, print)
```


Q7a vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
Q7a
Identify
Manage
Refer
Q7b vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q7b
Identify
Manage
Refer


Q7c vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q7c
Identify
Manage
Refer
Q7d vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
80
Q7d
Identify
Manage
Refer


Q8a vs Q2
No
Unsure
Yes
Percent of responses
60
40
20
0
Q8a
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree
Q8b vs Q2
No
Unsure
Yes
Percent of responses
60
40
20
0
Q8b
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree

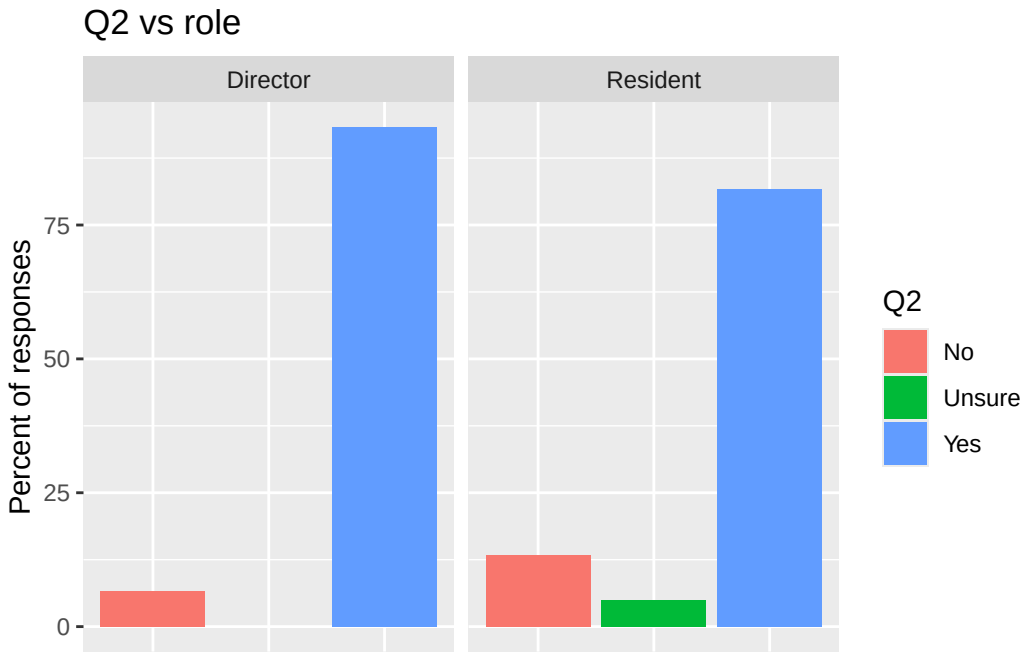

Q8c vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8c
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree
Q8d vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8d
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree

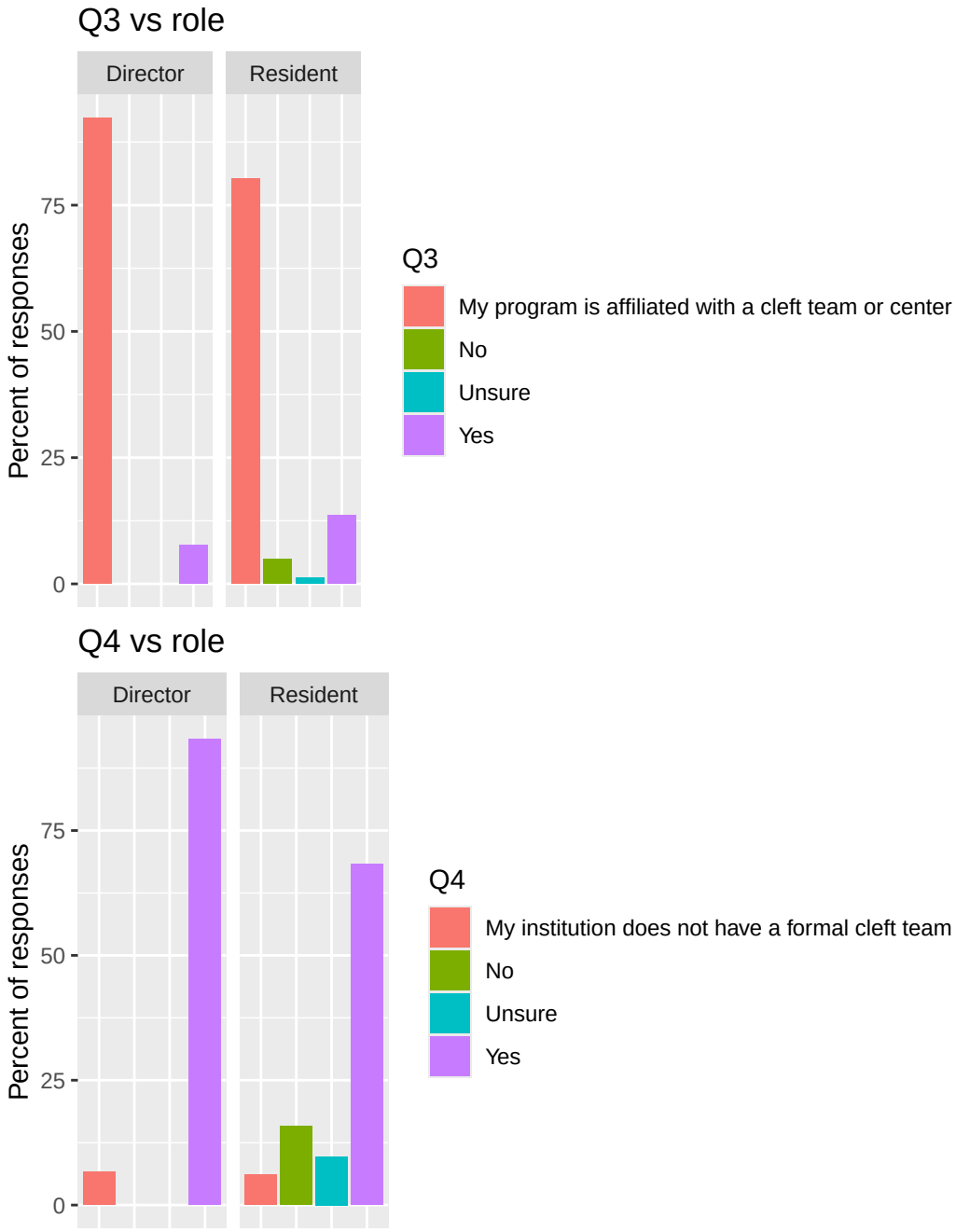


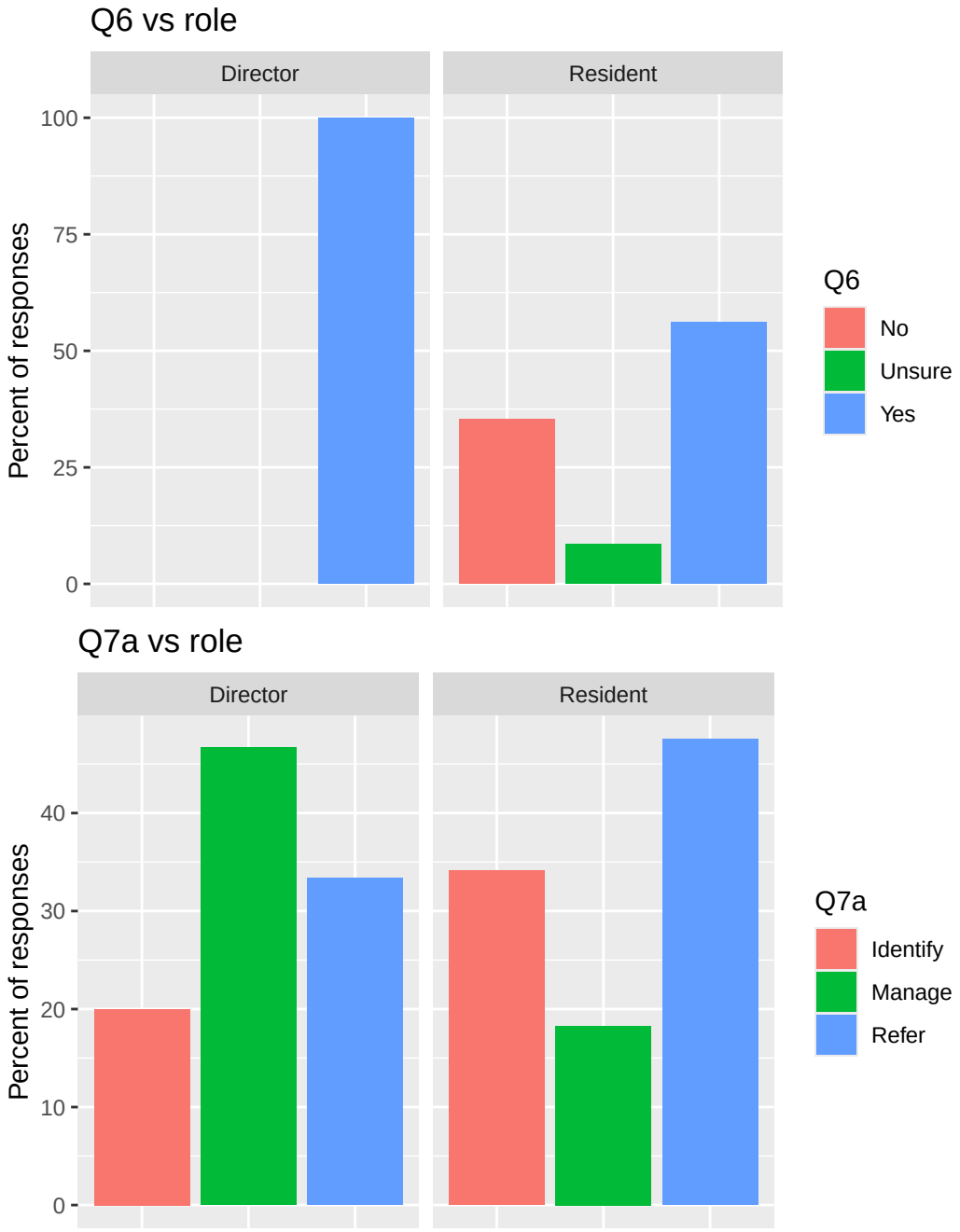

```
question_dic_res_dir <- data.frame(question1s_res_dir = colnames(res_dir)
                                   , question2s_res_dir = rep("role", ncol(res_dir)))

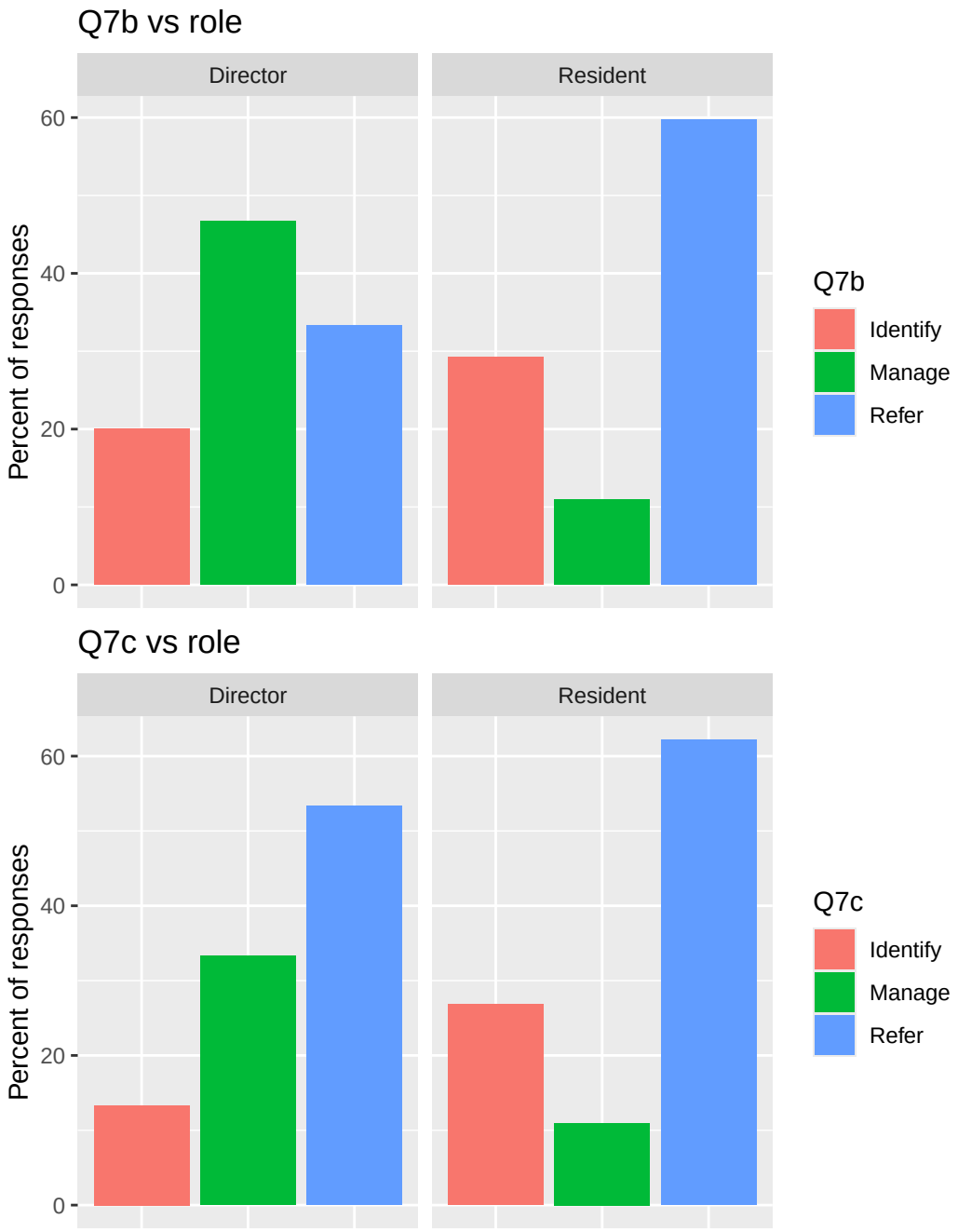
question_dic_res_dir <- question_dic_res_dir %>%
  filter(question1s_res_dir != "role")
```

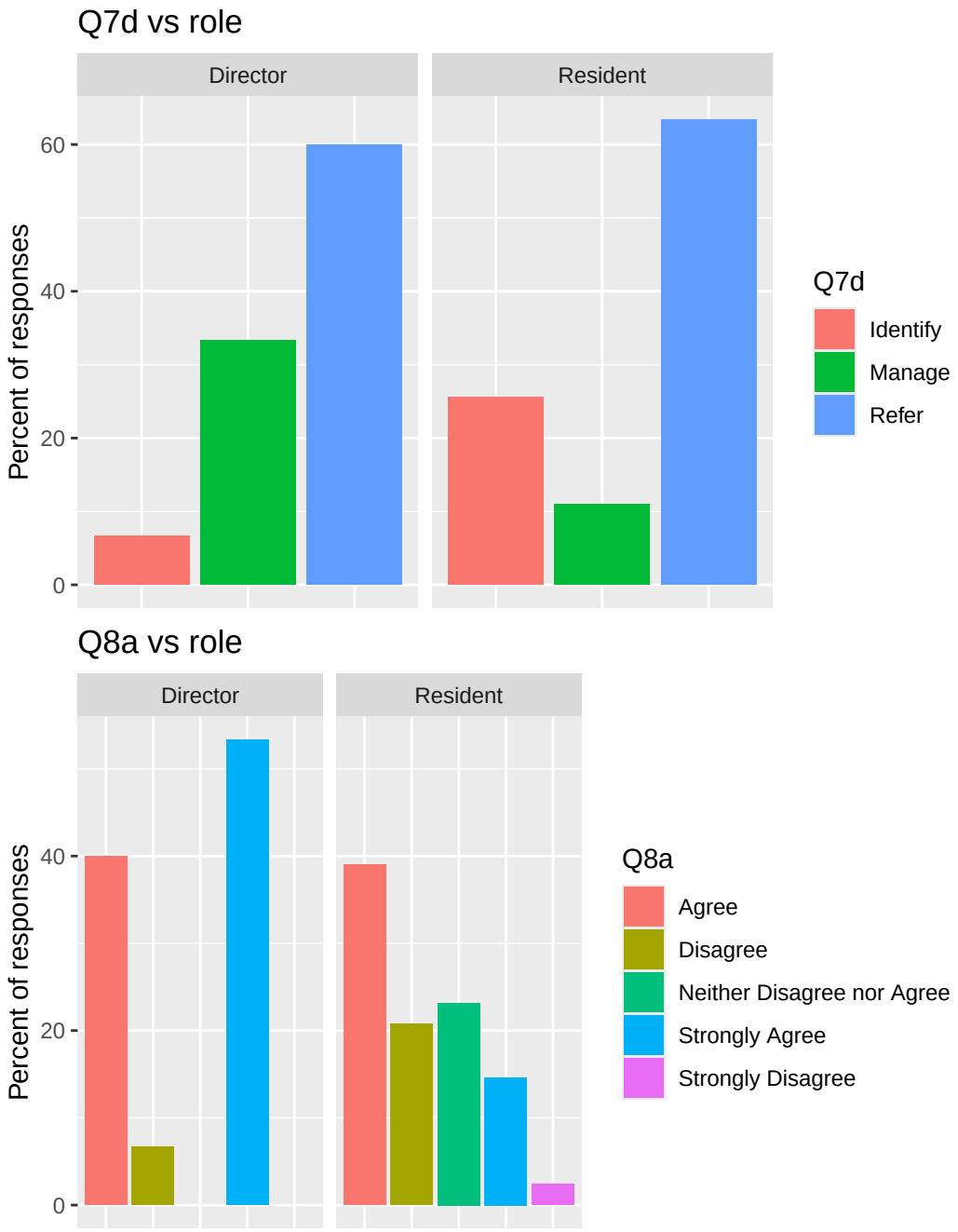
```r
plots_comp <- pmap(question_dic_res_dir,
                   function(question1s_res_dir, question2s_res_dir){
                     stack_bar(
                       q_reference = question1s_res_dir,
                       q_group = question2s_res_dir,
                       data = res_dir
                     )
                   })
walk(plots_comp, print)
```

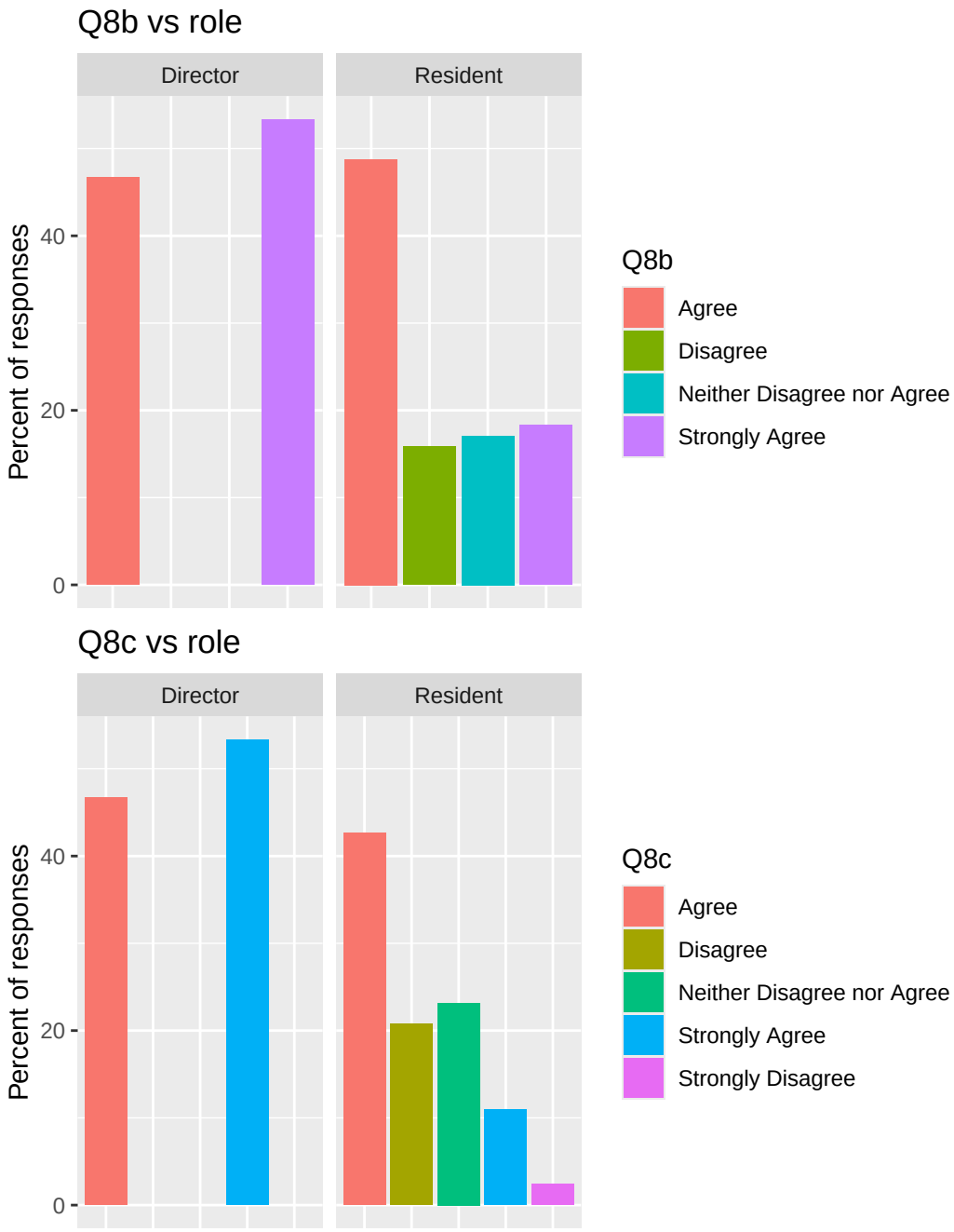


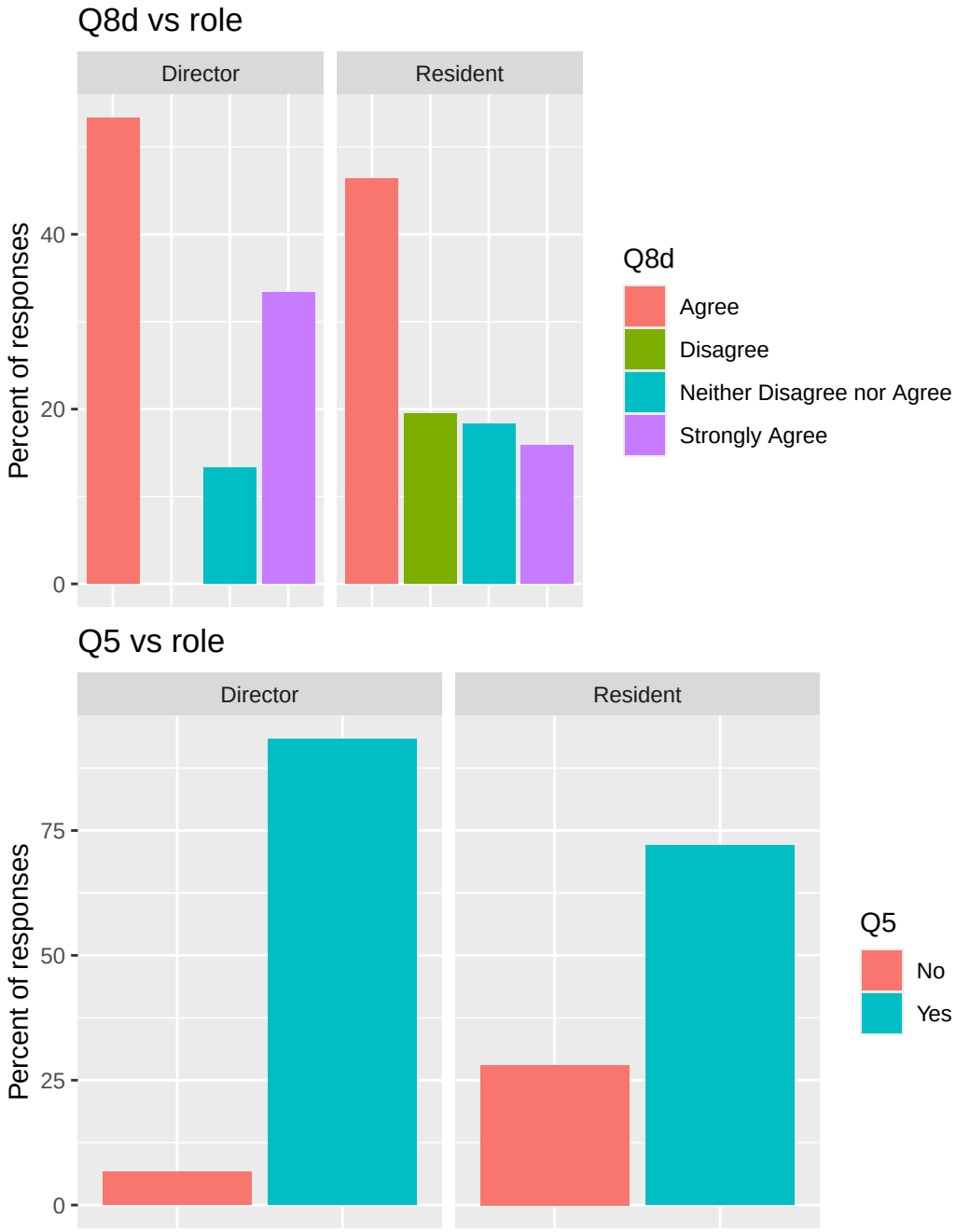

Q3 vs role
Director
Resident
Percent of responses
75
50
25
0
Q3
My program is affiliated with a cleft team or center
No
Unsure
Yes
Q4 vs role
Director
Resident
Percent of responses
75
50
25
0
Q4
My institution does not have a formal cleft team
No
Unsure
Yes

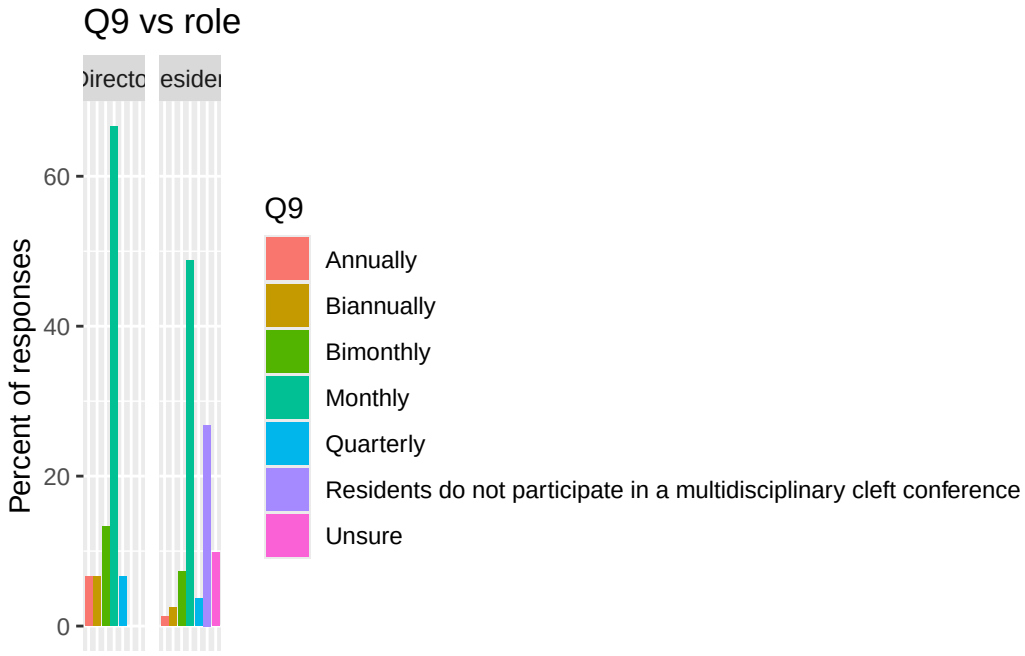

Q6 vs role
Director
Resident
Percent of responses
100
75
50
25
0
Q6
No
Unsure
Yes
Q7a vs role
Director
Resident
Percent of responses
40
30
20
10
0
Q7a
Identify
Manage
Refer

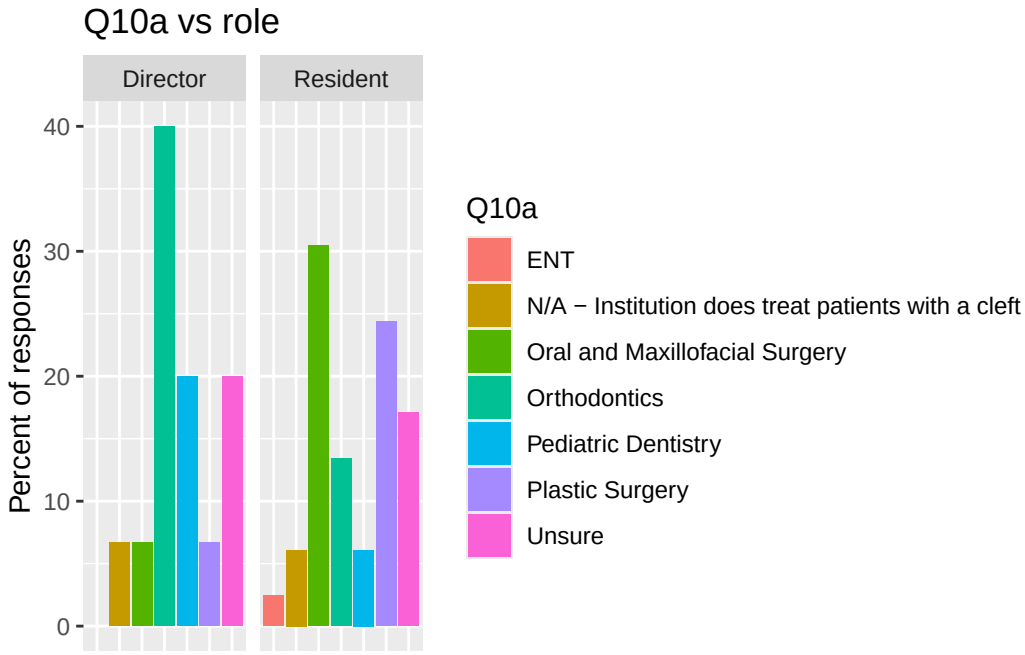

Q7b vs role
Director
Resident
Percent of responses
60
40
20
0
Q7b
Identify
Manage
Refer
Q7c vs role
Director
Resident
Percent of responses
60
40
20
0
Q7c
Identify
Manage
Refer


Q7d vs role
Director
Resident
Percent of responses
60
40
20
0
Q7d
Identify
Manage
Refer
Q8a vs role
Director
Resident
Percent of responses
40
20
0
Q8a
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

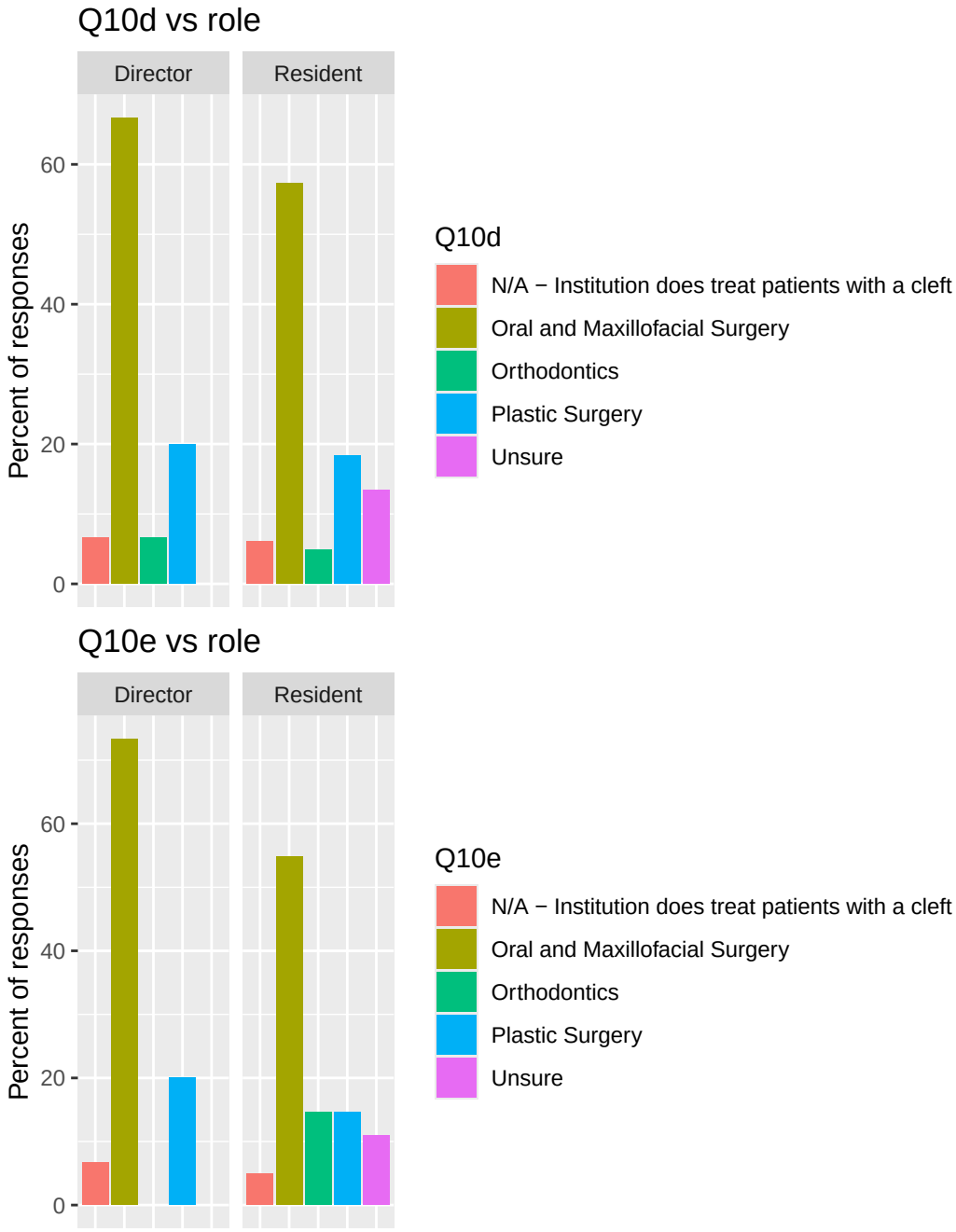

Q8b vs role
Director
Resident
Percent of responses
40
20
0
Q8b
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q8c vs role
Director
Resident
Percent of responses
40
20
0
Q8c
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

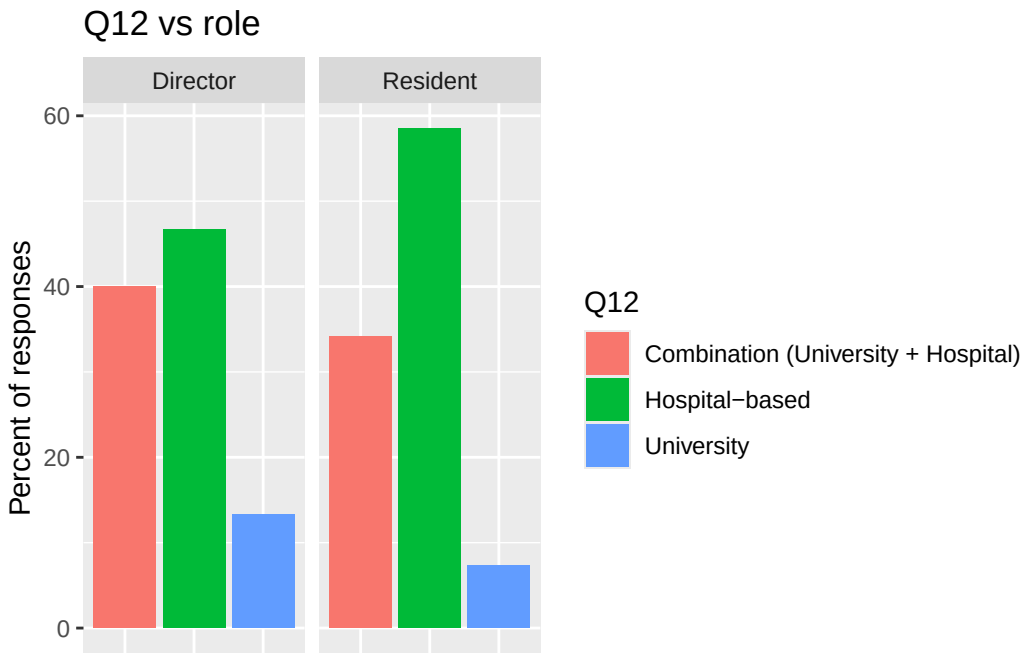

Q8d vs role
Director
Resident
Percent of responses
40
20
0
Q8d
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q5 vs role
Director
Resident
Percent of responses
75
50
25
0
Q5
No
Yes


Q9 vs role
Percent of responses
60
40
20
0
Q9
Annually
Biannually
Bimonthly
Monthly
Quarterly
Residents do not participate in a multidisciplinary cleft conference
Unsure


Q10a vs role
Director
Resident
Percent of responses
40
30
20
10
0
Q10a
ENT
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Pediatric Dentistry
Plastic Surgery
Unsure


Q10b vs role
Director
Resident
Percent of responses
60
40
20
0
Q10b
ENT
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Plastic Surgery
Unsure
Q10c vs role
Director
Resident
Percent of responses
40
20
0
Q10c
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Plastic Surgery
Unsure


Q10d vs role
Director
Resident
Percent of responses
60
40
20
0
Q10d
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Plastic Surgery
Unsure
Q10e vs role
Director
Resident
Percent of responses
60
40
20
0
Q10e
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Plastic Surgery
Unsure



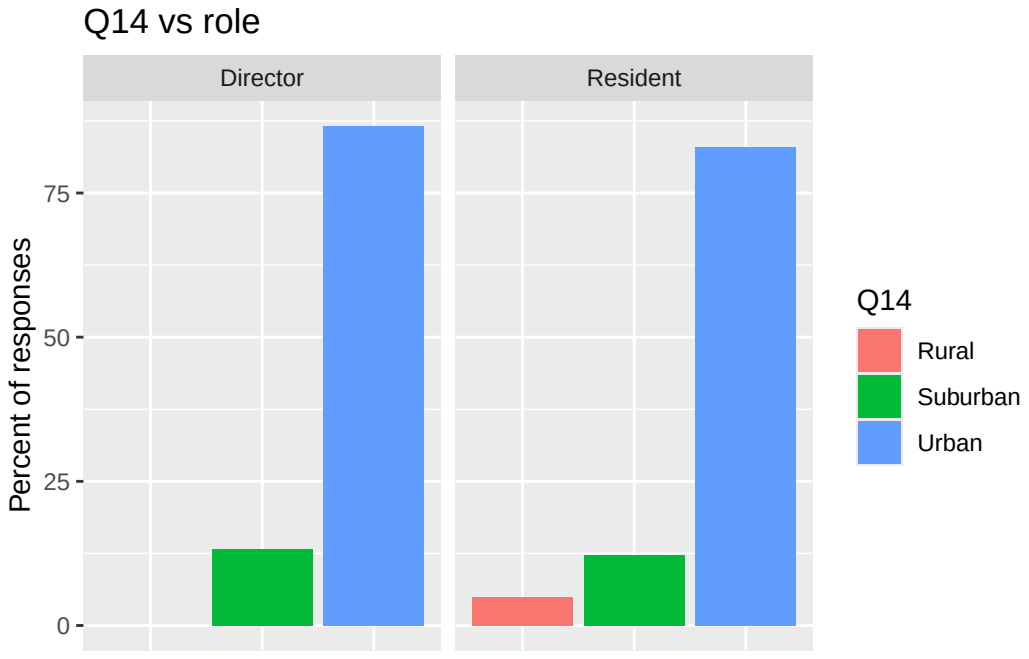
ı, Ohio, South Dakota, Wisconsin, and the Canadian provinces of Manitoba and Ontario)

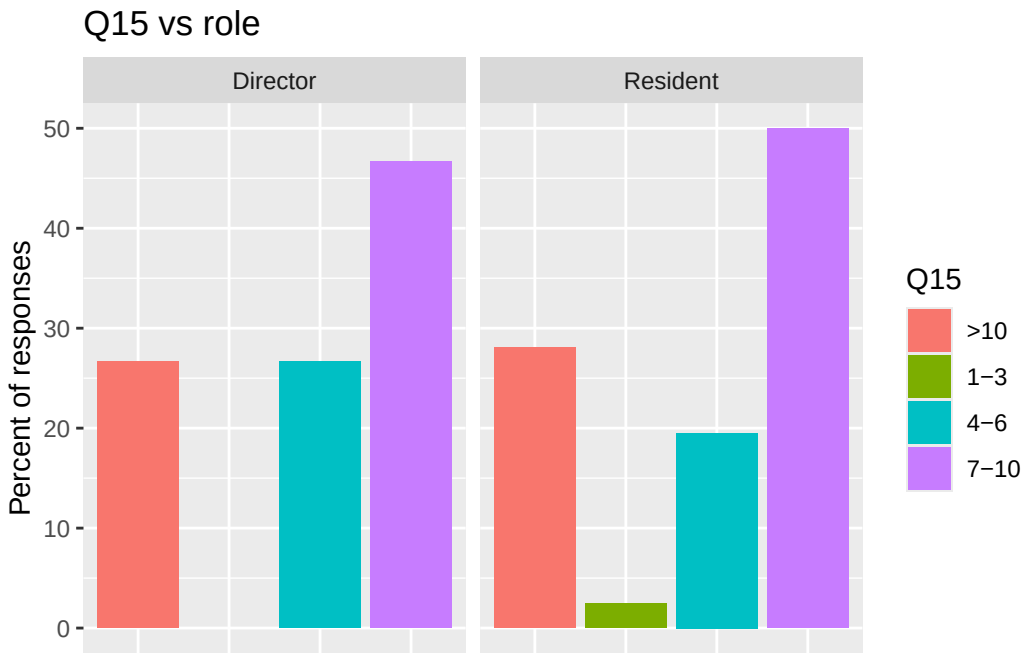
usetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and the Cana

ı Carolina, Tennessee, Virginia, West Virginia, the Commonwealth of Puerto Rico, and the US Virgin

and Mexico

Washington, Wyoming, and the Canadian provinces of Saskatchewan, Alberta, British Columbia, No




Q15 vs role
Director
Resident
Percent of responses
50
40
30
20
10
0
Q15
>10
1–3
4–6
7–10